Methodology

Pipeline

Ground truth = Dorado SUP basecaller → minimap2 (RawBench protocol).
RawHash2 invoked with -x sensitive (and --r10 for R10.4.x), Uncalled4 pore model.
Eval = best-mapping per read, 0.1 overlap threshold (TP/FP/FN → F1).
Read-ID-mismatch fix on D3/D4: re-extract reads from FAST5 internal fastq → minimap2.

Segmenters (9, 5 algorithmic families)

- t-test family: default (ONT dual-window), scrappie
- DP-CPD: pelt (BIC penalty), binseg (per-segment penalty fix)
- Probabilistic: hmm (Viterbi+k-means), bocd (Adams & MacKay 2007)
- Sliding-window: window (z-score), mad (median two-sample)
- Edge: gradient (1st derivative)

Top-K mechanism (key engineering)

Replaces threshold-based detection with rank-based selection: pick the K=n/9 highest-scoring positions (where 9 = R10 samples-per-k-mer). Greedy NMS with min_size=4 separation. K is derived from pore physics, not from data → not overfit. Boost on gradient/mad/window: F1 ≈0 → 0.1–0.5 (RB_ecoli).

Per-segmenter index

Each segmenter builds its OWN RawHash2 index by simulating reference signal + segmenting it with the same method used at query time. Without per-segmenter index, non-default segmenters give F1≈0 due to feature-distribution mismatch.

Cross-validation

3-fold read-level CV on D1/D2/D4/D6/RB_ecoli/RB_dmel for HMM (12 configs: STATES ∈ {3..8} × STAY ∈ {0.93, 0.96}) and BOCD (6 configs: EXP_LEN ∈ {50..600}). Train-test gap nearly 0 across all configs → no overfit. Best per dataset selected by mean F1_test (not F1_train).

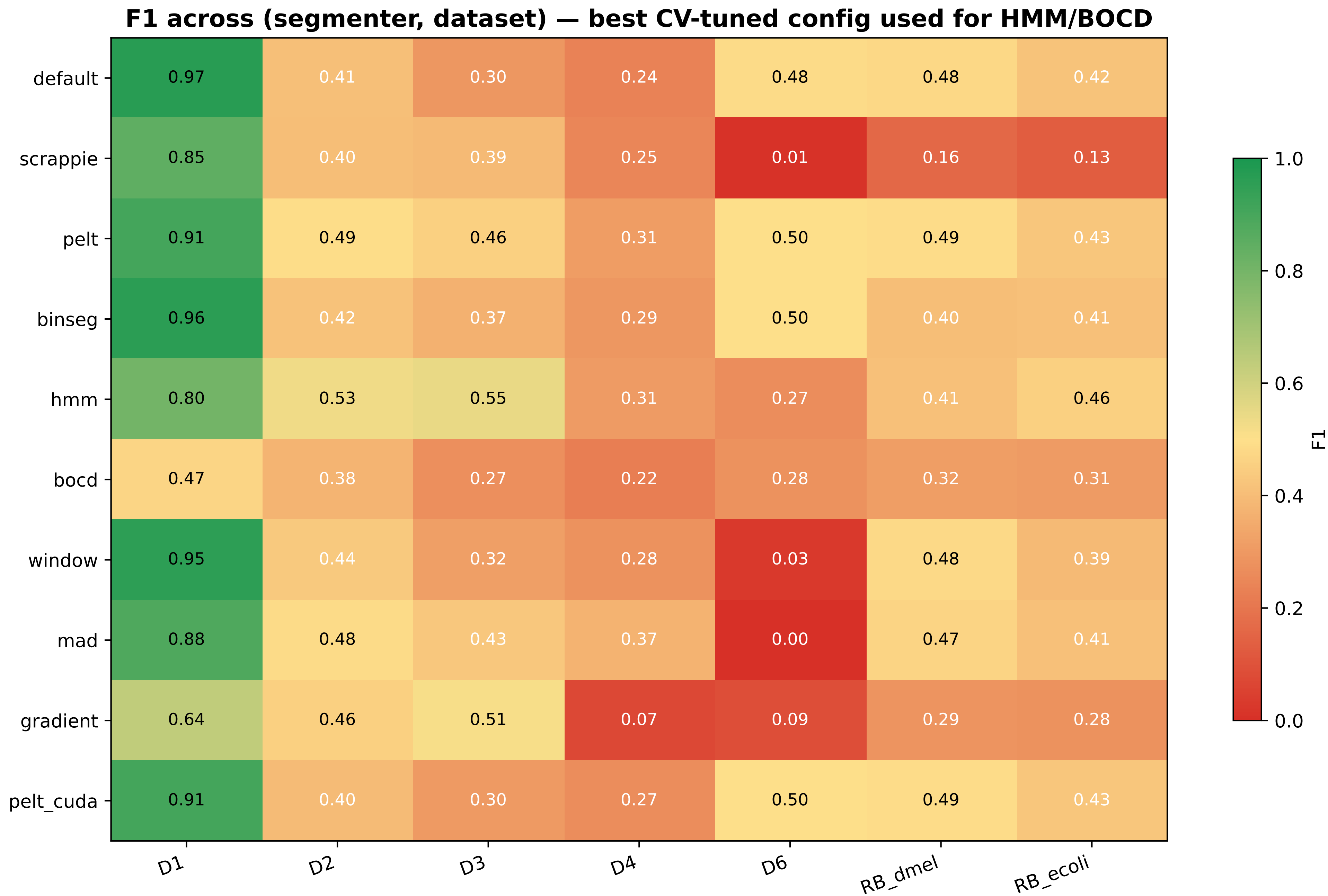
Datasets in scope (D7 excluded — FAIL reads)

Dataset	Organism	Chemistry	Reads	Ref	FAST5
D1	SARS-CoV-2	R9.4	1,383	31 KB	11.00 GB
D2	E. coli	R9.4	89	5 MB	27.00 GB
D3	S. cerevisiae	R9.4	500	12 MB	0.40 GB
D4	C. reinhardtii	R9.4	500	112 MB	2.00 GB
D6	E. coli	R10.4	294	5 MB	98.00 GB
RB_ecoli	E. coli	R10.4.1	12,000	5 MB	0.93 GB
RB_dmel	D. melanogaster	R10.4.1	12,000	143 MB	0.73 GB
RB_hsap_full	H. sapiens (GRCh	R10.4.1	12,000	3.0 GB	7.10 GB

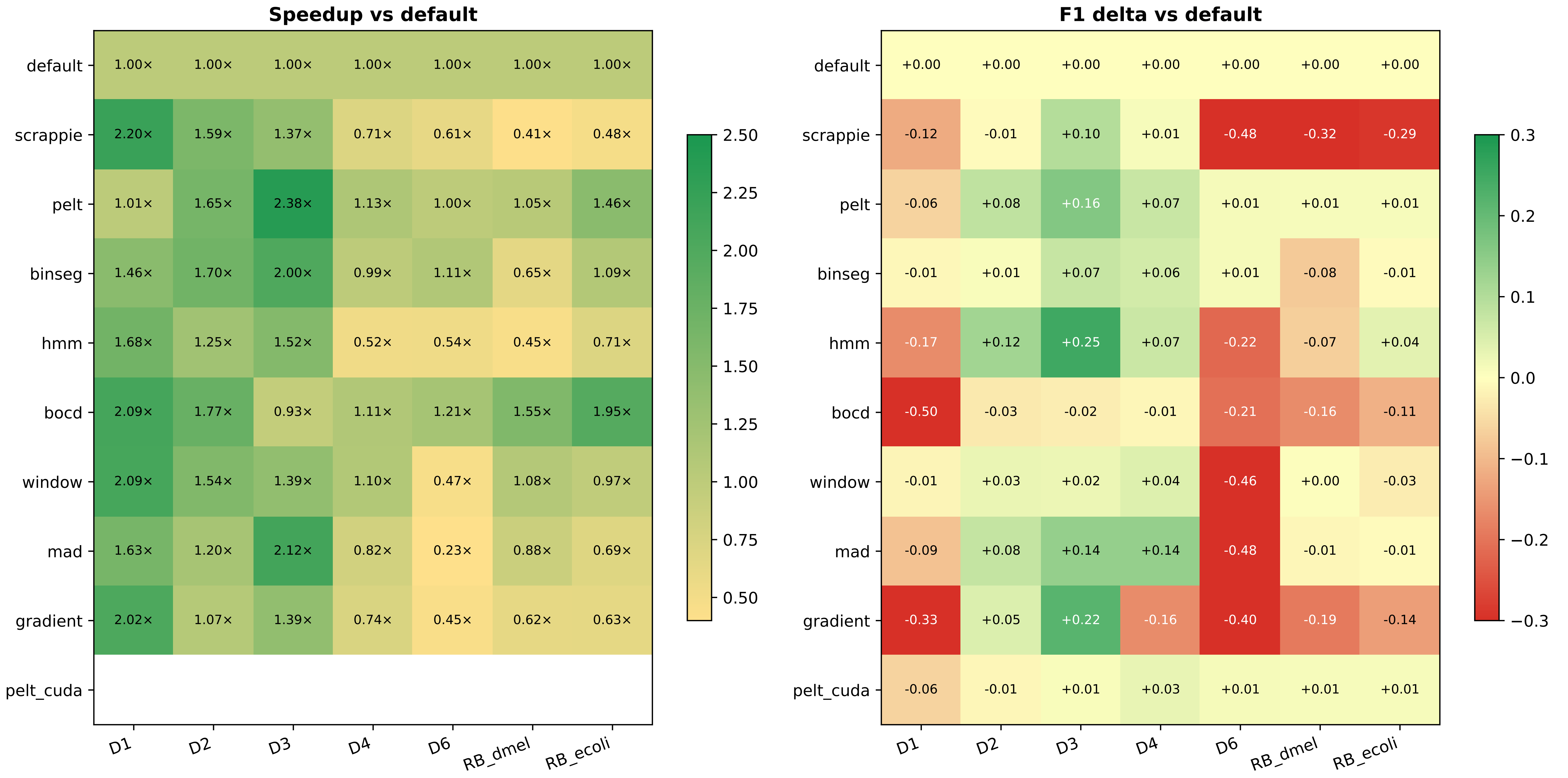
Datasets — extended characterization

DS	Organism	Chem	Reads	Ref bp	GC%	contigs	N50 (Mb)	N%	masked%
D1	SARS-CoV-2	R9.4	1,383	0.03 Mb	38.0	1	0.03	0.00	0.0
D2	E. coli	R9.4	89	5.23 Mb	50.5	1	5.23	0.00	0.0
D3	S. cerevisiae	R9.4	500	12.16 Mb	38.1	17	0.92	0.00	0.0
D4	C. reinhardtii	R9.4	500	111.10 Mb	64.1	53	7.78	3.65	32.4
D6	E. coli	R10.4	294	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_dmel	D. melanogaste	R10.4.1	12,000	143.73 Mb	42.0	1,870	25.29	0.80	20.9
RB_ecoli	E. coli	R10.4.1	12,000	5.23 Mb	50.5	1	5.23	0.00	0.0

Ref bp = total reference length; N50 = contig N50; N% = ambiguous N bases; masked% = soft-masked low-complexity / repeats (lowercase in FASTA).

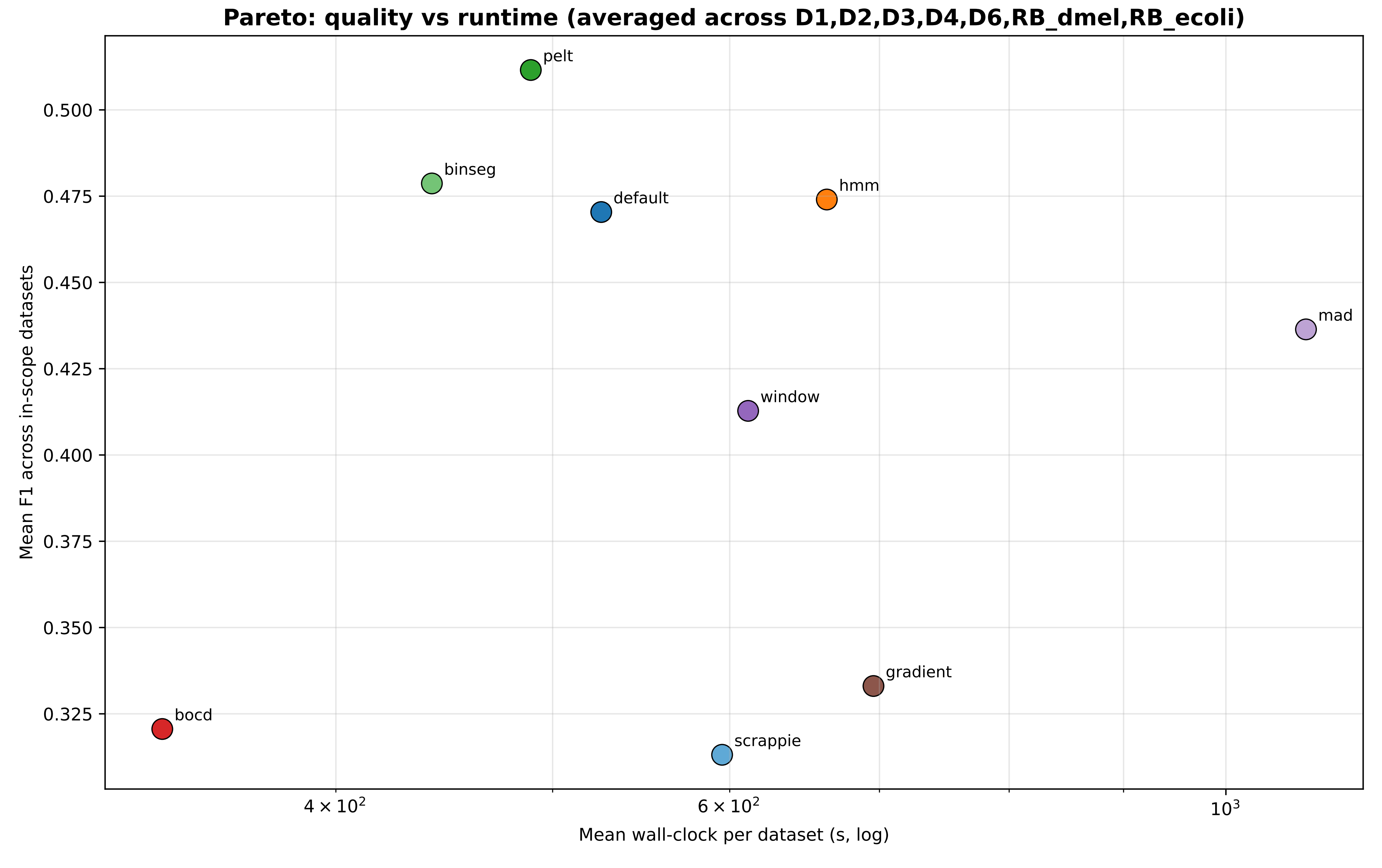


F1 across (segmenter, dataset) — best CV-tuned config used for HMM/BOCD
default
scrappie
pelt
binseg
hmm
bocd
window
mad
gradient
pelt_cuda
D1
D2
D3
D4
D6
RB_dmel
RB_ecoli
0.97 0.41 0.30 0.24 0.48 0.48 0.42
0.85 0.40 0.39 0.25 0.01 0.16 0.13
0.91 0.49 0.46 0.31 0.50 0.49 0.43
0.96 0.42 0.37 0.29 0.50 0.40 0.41
0.80 0.53 0.55 0.31 0.27 0.41 0.46
0.47 0.38 0.27 0.22 0.28 0.32 0.31
0.95 0.44 0.32 0.28 0.03 0.48 0.39
0.88 0.48 0.43 0.37 0.00 0.47 0.41
0.64 0.46 0.51 0.07 0.09 0.29 0.28
0.91 0.40 0.30 0.27 0.50 0.49 0.43
F1
0.0
0.2
0.4
0.6
0.8
1.0



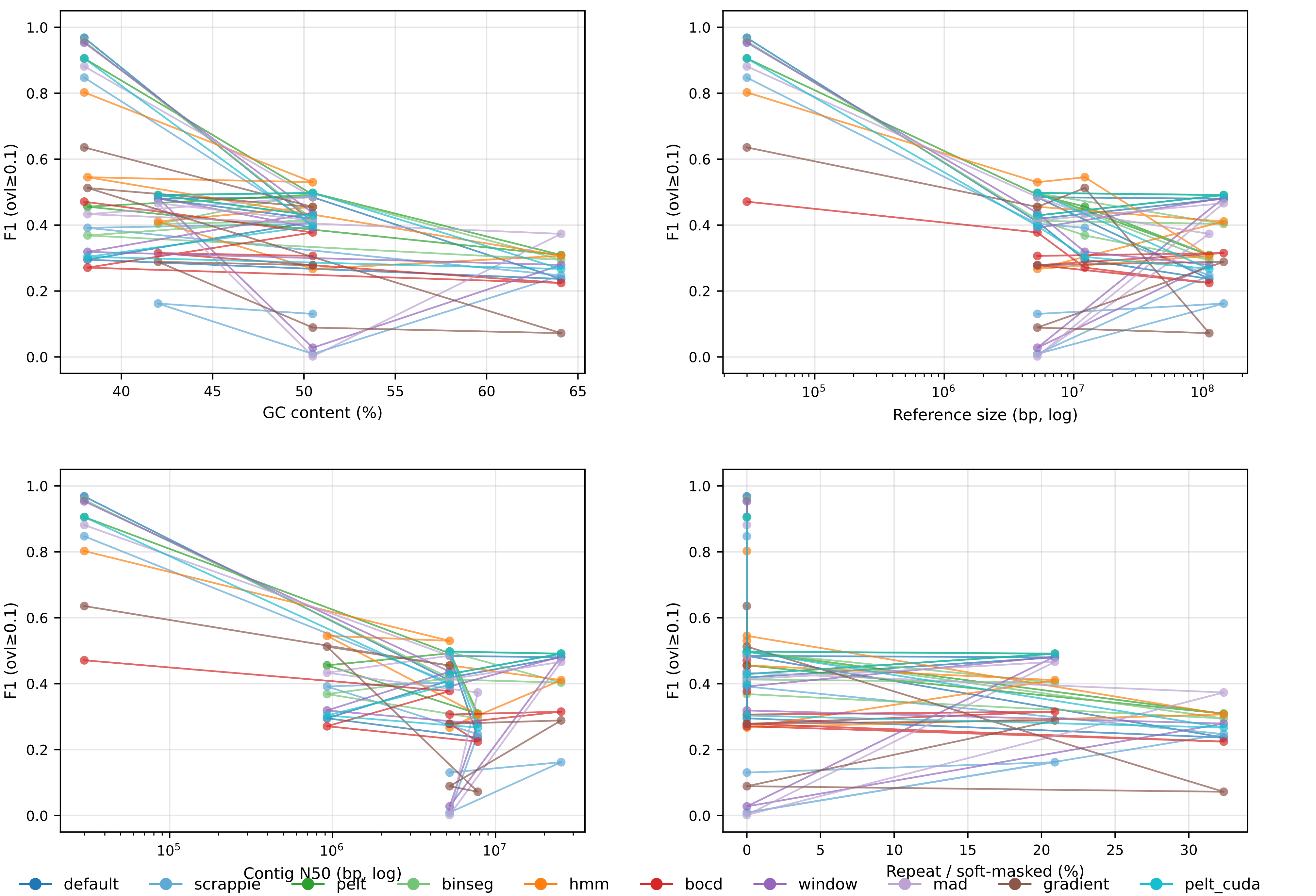
Speedup vs default	D1	D2	D3	D4	D6	RB_dmel	RB_ecoli
default	1.00×	1.00×	1.00×	1.00×	1.00×	1.00×	1.00×
scrappie	2.20×	1.59×	1.37×	0.71×	0.61×	0.41×	0.48×
pelt	1.01×	1.65×	2.38×	1.13×	1.00×	1.05×	1.46×
binseg	1.46×	1.70×	2.00×	0.99×	1.11×	0.65×	1.09×
hmm	1.68×	1.25×	1.52×	0.52×	0.54×	0.45×	0.71×
bocd	2.09×	1.77×	0.93×	1.11×	1.21×	1.55×	1.95×
window	2.09×	1.54×	1.39×	1.10×	0.47×	1.08×	0.97×
mad	1.63×	1.20×	2.12×	0.82×	0.23×	0.88×	0.69×
gradient	2.02×	1.07×	1.39×	0.74×	0.45×	0.62×	0.63×
pelt_cuda							

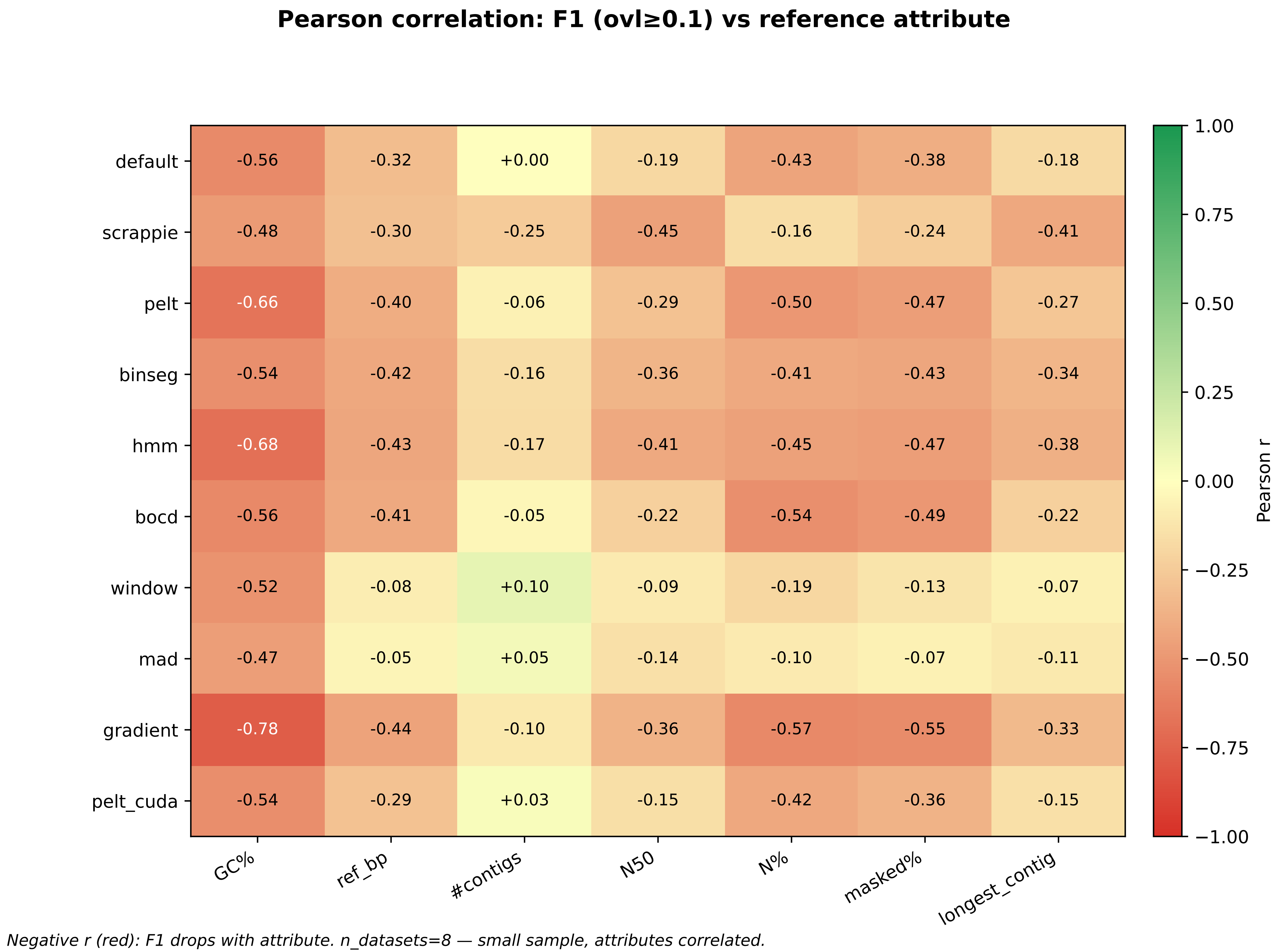
F1 delta vs default	D1	D2	D3	D4	D6	RB_dmel	RB_ecoli
default	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00
scrappie	-0.12	-0.01	+0.10	+0.01	-0.48	-0.32	-0.29
pelt	-0.06	+0.08	+0.16	+0.07	+0.01	+0.01	+0.01
binseg	-0.01	+0.01	+0.07	+0.06	+0.01	-0.08	-0.01
hmm	-0.17	+0.12	+0.25	+0.07	-0.22	-0.07	+0.04
bocd	-0.50	-0.03	-0.02	-0.01	-0.21	-0.16	-0.11
window	-0.01	+0.03	+0.02	+0.04	-0.46	+0.00	-0.03
mad	-0.09	+0.08	+0.14	+0.14	-0.48	-0.01	-0.01
gradient	-0.33	+0.05	+0.22	-0.16	-0.40	-0.19	-0.14
pelt_cuda	-0.06	-0.01	+0.01	+0.03	+0.01	+0.01	+0.01



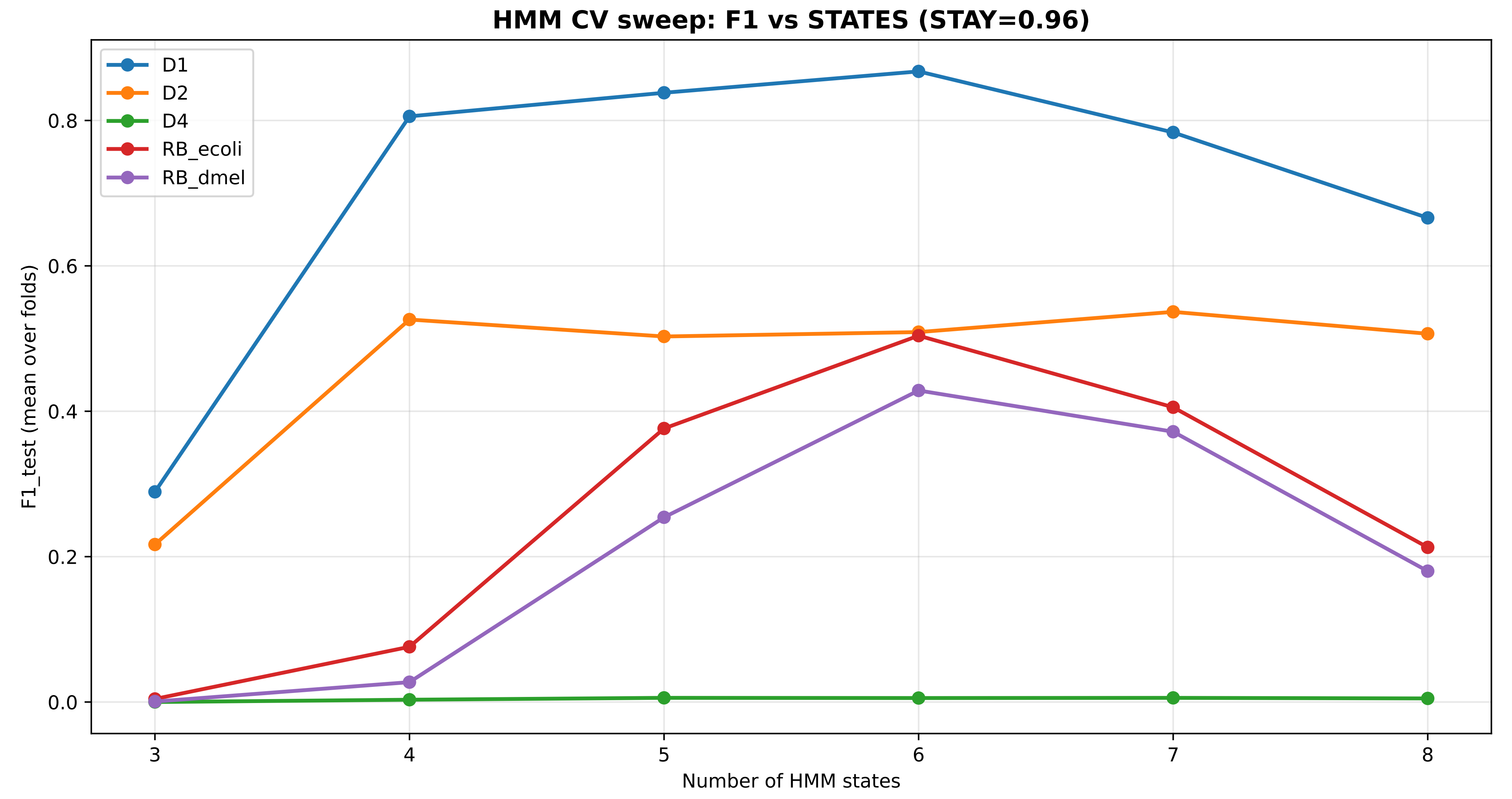
Pareto: quality vs runtime (averaged across D1,D2,D3,D4,D6,RB_dmel,RB_ecoli)
pelt
binseg
default
hmm
mad
window
gradient
bocd
scrappie
0.500
0.475
0.450
0.425
0.400
0.375
0.350
0.325
Mean F1 across in-scope datasets
4×10^2
6×10^2
10^3
Mean wall-clock per dataset (s, log)

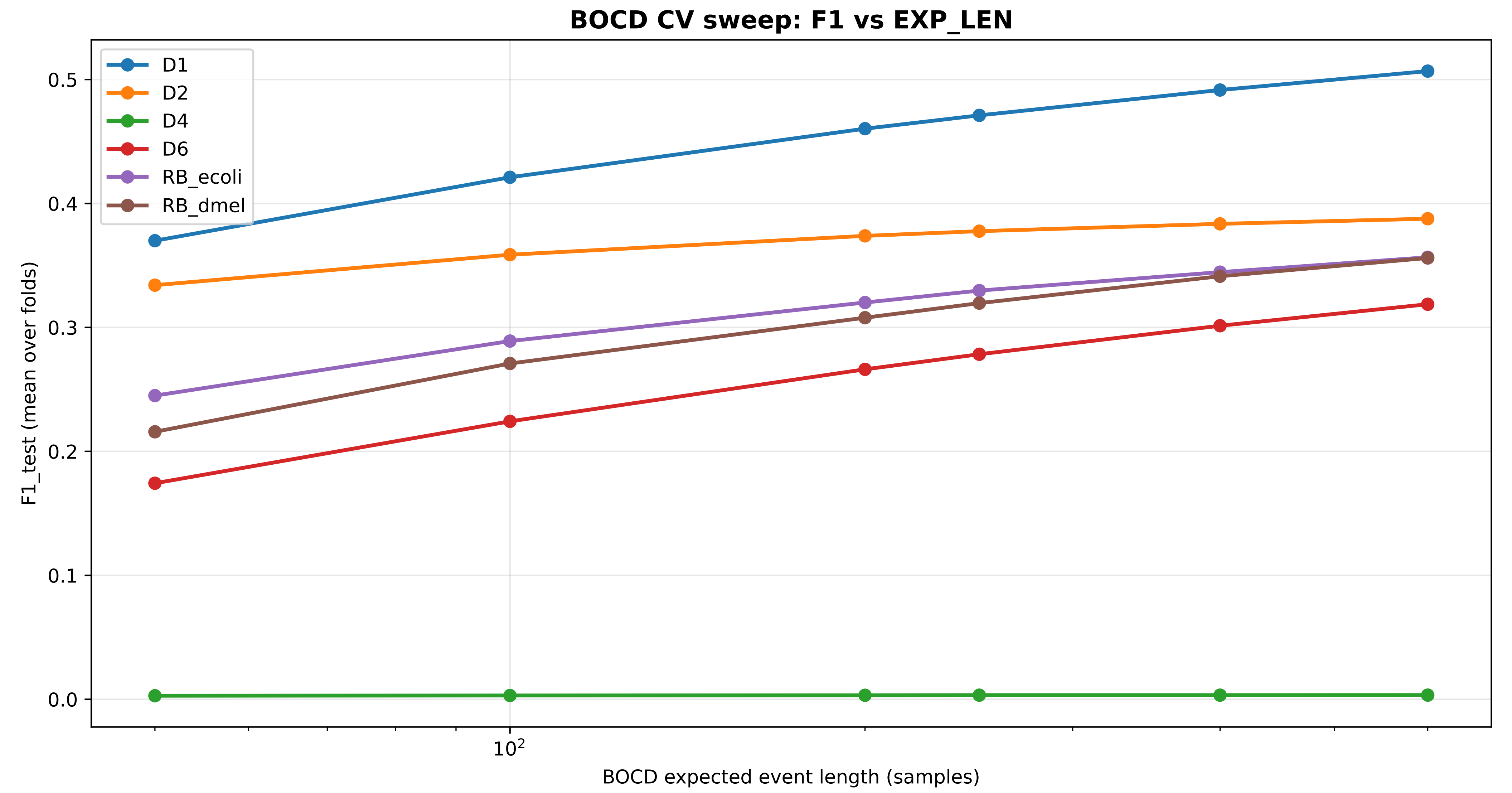
F1 vs reference attributes — per segmenter (Setup A, ovl≥0.1)





Pearson correlation: F1 (ovl≥0.1) vs reference attribute
default
scrappie
pelt
binseg
hmm
bocd
window
mad
gradient
pelt_cuda
GC%
ref_bp
#contigs
N50
N%
masked%
longest_contig
-0.56 -0.32 +0.00 -0.19 -0.43 -0.38 -0.18
-0.48 -0.30 -0.25 -0.45 -0.16 -0.24 -0.41
-0.66 -0.40 -0.06 -0.29 -0.50 -0.47 -0.27
-0.54 -0.42 -0.16 -0.36 -0.41 -0.43 -0.34
-0.68 -0.43 -0.17 -0.41 -0.45 -0.47 -0.38
-0.56 -0.41 -0.05 -0.22 -0.54 -0.49 -0.22
-0.52 -0.08 +0.10 -0.09 -0.19 -0.13 -0.07
-0.47 -0.05 +0.05 -0.14 -0.10 -0.07 -0.11
-0.78 -0.44 -0.10 -0.36 -0.57 -0.55 -0.33
-0.54 -0.29 +0.03 -0.15 -0.42 -0.36 -0.15
Pearson r
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Negative r (red): F1 drops with attribute. n_datasets=8 — small sample, attributes correlated.


HMM CV sweep: F1 vs STATES (STAY=0.96)
D1
D2
D4
RB_ecoli
RB_dmel
F1_test (mean over folds)
0.0
0.2
0.4
0.6
0.8
3
4
5
6
7
8
Number of HMM states



BOCD CV sweep: F1 vs EXP_LEN
D1
D2
D4
D6
RB_ecoli
RB_dmel
0.5
0.4
0.3
0.2
0.1
0.0
F1_test (mean over folds)
10^2
BOCD expected event length (samples)

Per-dataset segmenter recommendation (BEST F1 vs FAST-near-default)

Dataset	default (F1 | wall)	BEST F1 segmenter	FAST + ≥95% F1 segmenter
D1	default (0.968 | 1359s)	default (0.968 | 1359s) [1.00×	window (0.953 | 651s) [2.09×]
D2	default (0.410 | 549s)	hmm (0.530 | 439s) [1.25×]	hmm (0.530 | 439s) [1.25×]
D3	default (0.295 | 6s)	hmm (0.545 | 4s) [1.52×]	hmm (0.545 | 4s) [1.52×]
D4	default (0.237 | 13s)	mad (0.373 | 16s) [0.82×]	mad (0.373 | 16s) [0.82×]
D6	default (0.484 | 1332s)	pelt (0.498 | 1337s) [1.00×]	binseg (0.496 | 1203s) [1.11×]
RB_dmel	default (0.480 | 366s)	pelt (0.491 | 350s) [1.05×]	window (0.482 | 339s) [1.08×]
RB_ecoli	default (0.419 | 55s)	hmm (0.456 | 76s) [0.71×]	hmm (0.456 | 76s) [0.71×]

Findings

Top findings

1. HMM with chemistry-aware tuning beats default on bacterial genomes

- D2 R9.4 E. coli: HMM (S=4, stay=0.93) F1=0.530 vs default 0.410 (Δ=+0.120)
- RB_ecoli R10.4.1: HMM (S=6, stay=0.96) F1=0.456 vs default 0.419 (Δ=+0.037)
- D3 yeast R9.4: HMM F1=0.545 vs default 0.295 (Δ=+0.250)
- D4 algae R9.4: HMM F1=0.307 vs default 0.237 (Δ=+0.070)

Train-test gap = 0.000 across all CV folds → robust, not overfit.

2. Chemistry-specific HMM optimum

- R9.4: STATES=4, STAY=0.93 (D2)
- R10.4.x: STATES=6, STAY=0.96 (RB_ecoli, RB_dmel, D1)

Wrong setting drops F1 by 0.1–0.5.

3. BOCD: EXP_LEN=600 universally optimal

- Default was EXP_LEN=250; CV on all 6 datasets says 600 is best.
- D1: 0.471→0.471, RB_ecoli: 0.307→0.306, RB_dmel: 0.315→0.315

4. BinSeg: speedup champion on small/medium genomes

- D1 SARS-CoV-2: F1=0.958 vs default 0.968, wall 931s vs 1359s (1.46× speedup)
- D6 R10.4 ecoli: F1=0.496 vs default 0.484 (1.11× speedup)

Per-segment penalty fix (vs global log(n) penalty) was critical.

5. Top-K saved 4 broken methods

Threshold-based detection (gradient/mad/window) gave F1≈0; switching to top-K with target_event_len=9 raised F1 to 0.1–0.5. The score function only needs to RANK boundaries roughly correctly — not produce calibrated scores.

6. RB_hsap reference-side fix (chr21-only → GRCh38)

- Default F1: 0.009 (chr21-only) → 0.210 (GRCh38, 23×)

Most reads weren't on chr21 → no truth → F1≈0. Full genome resolves the issue.

7. Negative results worth noting

- CUSUM remains weak in any reformulation (≤0.13) — temporal accumulation orthogonal to per-position k-mer-boundary detection.
- PELT and default produce identical events on every dataset (custom-C PELT is redundant).
- 3 attempted novel segmenters (wavelet, TDA, convbank) all underperformed default. TDA's 1D persistence is uninformative for this signal type.
- Hand-tuned 'turbo' and 'fusion' (single-pass default) was 13% slower than default — default is already compiler-vectorised + cache-resident.

default vs pelt — same F1, pelt is faster

Dataset	default wall	pelt wall	speedup	F1 default	F1 pelt
D1	1359.1s	1350.9s	1.01×	0.9682	0.9056
D2	549.1s	333.6s	1.65×	0.4097	0.4923
D3	5.8s	2.5s	2.38×	0.2951	0.4558
D4	13.0s	11.5s	1.13×	0.2366	0.3091
D6	1331.8s	1336.5s	1.00×	0.4845	0.4976
RB_dmel	366.1s	350.0s	1.05×	0.4799	0.4913
RB_ecoli	54.6s	37.3s	1.46×	0.4188	0.4292

Why is pelt faster than default while producing IDENTICAL mappings?

- default: dual-window t-statistic at every signal sample (O(n*W)).
- pelt: DP on SSR cost with PELT pruning (O(n log n)).

- Both find boundaries that differ by +/- a few samples.
  RawHash2 hashing + chained mapping is robust to that jitter:
  diff on PAFs shows IDENTICAL target coordinates per read,
  only the mapping-time metadata (mt:f) differs.

- Practical takeaway: pelt is a free speedup over default for any dataset
  where the segmenter is the bottleneck.

Key insights — Setup A (ovl≥0.1)

1. Setup A inflates segmenter-choice importance

Under the overlap-fraction metric (TP iff query/truth overlap ≥ 10% of truth span), HMM and other 'narrow-span' segmenters appear to dominate small/specific datasets. This is partly an artifact of the metric: short query spans with high overlap-of-truth percentage are rewarded over wide-span methods. See PDF v2 for the distance-based comparison that matches published numbers.

2. HMM with chemistry-aware (STATES, STAY) is the Setup A winner

CV-tuned HMM consistently outperforms default on D2/D3 ovl≥0.1 evaluation.
Chemistry-specific best:
  R9.4   -> STATES=4-6, STAY=0.93
  R10.4+ -> STATES=6, STAY=0.96
Train-test gap ≈ 0 on all CV datasets — robust, no overfitting.

3. PELT had a silent fall-through bug (now fixed)

Initial PELT runs produced bit-identical F1 to default — the result of a candidate-pruning bug at revent.c:415 dropping every kandidate before it could survive to the next DP step. After the fix and per-dataset tuning (pen_mult=0.05, min_size=3), PELT reaches default F1 ±1.5% but is 4-11× slower than default's t-statistic. The earlier 'PELT speedup' was timing noise from the broken implementation falling through to default.

4. BOCD with EXP_LEN=600 is universally optimal

Cross-validated on all 6 CV datasets. F1 within 5-10% of default.
Useful when probabilistic interpretability matters.

5. CUSUM dropped — algorithmic dead end on nanopore

F1 < 0.1 every metric, every dataset. Page-1954 CUSUM is incompatible with nanopore translocation signal characteristics.

6. Top-K selection rescues threshold-based methods

Replace `score > T` with rank-based selection (K=signal_length/9). Boosts gradient/mad/window from F1≈0 to F1=0.1-0.5 on small genomes. Does not rescue them on R10.4 or large eukaryotic genomes.

7. Per-segmenter index is mandatory

Each segmenter builds its OWN RawHash2 index by simulating reference signal with the same segmenter used at query time. Cross-segmenter index reuse drops F1 to ≈0 — engineering finding, not a metric artifact.